

Chapter 6:

4. If $X \sim \chi_{10}^2$, find the constants a and b so that $\mathbb{P}(a < X < b) = 0.90$ and $\mathbb{P}(X < a) = 0.05$.

Solution:

$X \sim \chi_{10}^2$, $\mathbb{P}(a < X < b) = 0.90$, and $\mathbb{P}(X < a) = 0.05 \implies a = 3.9403$, and $b = 18.307$.

```
> a <- qchisq(0.05, 10)
> b <- qchisq(0.95, 10)
> c(a, b)

[1] 3.940299 18.307038
```

11. Given a random sample of size 6 from $N(0, \sigma)$, calculate

(a) $\mathbb{P}\left(\frac{\bar{X}}{S} > 2\right)$ and

(b) $\mathbb{P}\left(\left|\frac{\bar{X}}{S_u}\right| \leq 4\right)$.

Solution:

Note that $\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$.

(a) $\mathbb{P}\left(\frac{\bar{X}}{S} > 2\right) = \mathbb{P}(t_5 > 2\sqrt{6}) = 0.0022$

```
> 1 - pt(sqrt(6)*2, 5)

[1] 0.002239216
```

(b) $\frac{\bar{X} - 0}{\frac{S_u \sqrt{\frac{n}{n-1}}}{\sqrt{n}}} = \frac{\bar{X}}{\frac{S_u}{\sqrt{n-1}}} = \frac{\bar{X} \sqrt{n-1}}{S_u} \sim t_{n-1}$.

$\mathbb{P}\left(\left|\frac{\bar{X}}{S_u}\right| \leq 4\right) = \mathbb{P}\left(-4\sqrt{5} \leq \frac{\bar{X}}{S_u} \sqrt{5} \leq 4\sqrt{5}\right) = \mathbb{P}\left(-4\sqrt{5} \leq t_5 \leq 4\sqrt{5}\right) = 0.9997$.

```
> pt(4*sqrt(5), 5) - pt(-4*sqrt(5), 5)

[1] 0.9997089
```

Chapter 7:

14. The following random samples $\mathbf{X}=(x_1, \dots, x_7)$ and $\mathbf{Y}=(y_1, \dots, y_{10})$ are drawn from $Pois(\lambda)$ and $Pois(2\lambda)$, respectively:

$\mathbf{X} \sim \text{Pois}(\lambda)$	4	2	5	7	3	4	3			
$\mathbf{Y} \sim \text{Pois}(2\lambda)$	6	10	1	6	3	5	5	4	7	5

- (a) Derive the maximum likelihood estimator of λ and calculate its variance.
 (b) Compute the maximum likelihood estimate of λ and its variance using the two random samples given.

Solution:

- (a) The maximum likelihood estimator of λ is

$$\begin{aligned}
 L(\lambda | \mathbf{x}, \mathbf{y}) &= L_1(\lambda | \mathbf{x}) L_2(\lambda | \mathbf{y}) \\
 L(\lambda | \mathbf{x}, \mathbf{y}) &= \prod_{i=1}^{n_1} e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} \cdot \prod_{i=1}^{n_2} e^{-2\lambda} \frac{(2\lambda)^{y_i}}{y_i!} \\
 \ln L(\lambda | \mathbf{x}, \mathbf{y}) &= \ln \left[e^{-n_1 \lambda} \prod_{i=1}^{n_1} \frac{\lambda^{x_i}}{x_i!} \cdot e^{-2n_2 \lambda} \prod_{i=1}^{n_2} \frac{(2\lambda)^{y_i}}{y_i!} \right] \\
 &= -n_1 \lambda + \ln(\lambda) \sum_{i=1}^{n_1} x_i - \sum_{i=1}^{n_1} \ln(x_i!) - 2n_2 \lambda + \ln(2\lambda) \sum_{i=1}^{n_2} y_i - \sum_{i=1}^{n_2} \ln(y_i!) \\
 \frac{\partial \ln L(\lambda | \mathbf{x}, \mathbf{y})}{\partial \lambda} &= -n_1 + \frac{\sum_{i=1}^{n_1} x_i}{\lambda} - 2n_2 + \frac{\sum_{i=1}^{n_2} y_i}{\lambda} \stackrel{\text{set}}{=} 0 \\
 \frac{\sum_{i=1}^{n_1} x_i}{\lambda} + \frac{\sum_{i=1}^{n_2} y_i}{\lambda} &= n_1 + 2n_2 \\
 \lambda &= \frac{\sum_{i=1}^{n_1} x_i + \sum_{i=1}^{n_2} y_i}{n_1 + 2n_2} \\
 \Rightarrow \hat{\lambda}(\mathbf{X}, \mathbf{Y}) &= \frac{\sum_{i=1}^{n_1} X_i + \sum_{i=1}^{n_2} Y_i}{n_1 + 2n_2}
 \end{aligned}$$

The variance of the maximum likelihood estimator is

$$\begin{aligned}
 Var[\hat{\lambda}(\mathbf{X}, \mathbf{Y})] &= \frac{Var[\sum_{i=1}^{n_1} X_i + \sum_{i=1}^{n_2} Y_i]}{(n_1 + 2n_2)^2} \\
 &= \frac{n_1 Var[X] + n_2 Var[Y]}{(n_1 + 2n_2)^2} \\
 &= \frac{n_1 \lambda + 2n_2 \lambda}{(n_1 + 2n_2)^2} \\
 Var[\hat{\lambda}(\mathbf{X}, \mathbf{Y})] &= \frac{\lambda}{n_1 + 2n_2}
 \end{aligned}$$

- (b) The mle and its variance from the samples given are

$$\begin{aligned}
 \hat{\lambda}(\mathbf{x}, \mathbf{y}) &= \frac{\sum_{i=1}^{n_1} x_i + \sum_{i=1}^{n_2} y_i}{n_1 + 2n_2} = \frac{28 + 52}{7 + 2(10)} = 2.963 \\
 \widehat{Var}[\hat{\lambda}(\mathbf{x}, \mathbf{y})] &= \frac{\hat{\lambda}}{n_1 + 2n_2} = \frac{2.963}{7 + 2(10)} = 0.1097
 \end{aligned}$$

31. Consider the density function

$$f(x) = \frac{1}{\theta} x^{\frac{1-\theta}{\theta}}, \quad 0 < x < 1, \quad 0 < \theta < \infty.$$

- (a) Derive the maximum likelihood estimator of θ for a random sample of size n .
- (b) Derive the method of moments estimator of θ for a random sample of size n .
- (c) Show that the maximum likelihood estimator is unbiased.

Solution:

- (a) Deriving the maximum likelihood estimator of θ :

$$\begin{aligned} L(\theta|\mathbf{x}) &= \prod_{i=1}^n \frac{1}{\theta} x_i^{\frac{1-\theta}{\theta}} \\ &= \frac{1}{\theta^n} \prod_{i=1}^n x_i^{\frac{1-\theta}{\theta}} \\ \ln L(\theta|\mathbf{x}) &= -n \ln \theta + \frac{1-\theta}{\theta} \sum_{i=1}^n \ln(x_i) \\ \frac{\partial \ln L(\theta|\mathbf{x})}{\partial \theta} &= -\frac{n}{\theta} + \frac{(-1) \cdot \theta - 1(1-\theta)}{\theta^2} \sum_{i=1}^n \ln(x_i) \stackrel{\text{set}}{=} 0 \\ -\frac{n}{\theta} + \frac{(-1) \cdot \theta - 1(1-\theta)}{\theta^2} \sum_{i=1}^n \ln(x_i) &= 0 \\ \frac{-\theta - 1 + \theta}{\theta^2} \sum_{i=1}^n \ln(x_i) &= \frac{n}{\theta} \\ -\sum_{i=1}^n \ln(x_i) &= n\theta \\ \implies \theta &= \frac{-\sum_{i=1}^n \ln(x_i)}{n} \\ \implies \hat{\theta}(\mathbf{X}) &= \frac{-\sum_{i=1}^n \ln(X_i)}{n} \end{aligned}$$

(b) Deriving the method of moments estimator of θ :

$$\begin{aligned}
 \alpha_1(\theta) &= E[X] \stackrel{\text{set}}{=} \bar{X} = m_1 \\
 \int_0^1 x \cdot \frac{1}{\theta} x^{\frac{1-\theta}{\theta}} dx &= \bar{X} \\
 \frac{1}{\theta} \int_0^1 x^{\frac{1}{\theta}} &= \bar{X} \\
 \frac{1}{\theta} \cdot \frac{\theta}{1+\theta} x^{\frac{1+\theta}{\theta}} \Big|_0^1 &= \bar{X} \\
 \frac{1}{1+\theta} &= \bar{X} \\
 \frac{1}{\bar{X}} &= 1+\theta \\
 \theta &= \frac{1}{\bar{X}} - 1 \\
 \Rightarrow \hat{\theta}(\mathbf{X}) &= \frac{1 - \bar{X}}{\bar{X}}
 \end{aligned}$$

(c) If $E[\hat{\theta}(\mathbf{X})] = \theta$, the MLE is unbiased.

$$\begin{aligned}
 E[\hat{\theta}(\mathbf{X})] &= E\left[\frac{-\sum_{i=1}^n \ln(X_i)}{n}\right] \\
 &= \frac{-1}{n} \sum_{i=1}^n E[\ln(X_i)] \\
 &= -1E[\ln(X)]
 \end{aligned}$$

Recall that to integrate by parts, $\int_a^b u dv = uv|_a^b - \int_a^b v du$.
Calculating $E[\ln(X)]$:

$$E[\ln(X)] = \int_0^1 \ln(x) \frac{1}{\theta} x^{\frac{1-\theta}{\theta}} dx$$

Let $u = \ln(x)$ and $dv = \frac{1}{\theta} x^{\frac{1-\theta}{\theta}} dx$, so $du = \frac{1}{x} dx$ and $v = x^{\frac{1}{\theta}}$

$$\begin{aligned}
 &= \ln(x) \cdot x^{\frac{1}{\theta}} \Big|_0^1 - \int_0^1 x^{\frac{1}{\theta}} \cdot \frac{1}{x} dx \\
 &= 0 - \theta x^{\frac{1}{\theta}} \Big|_0^1 = -\theta
 \end{aligned}$$

So, $E[\hat{\theta}(\mathbf{X})] = -1E[\ln(X)] = -1(-\theta) = \theta$, and the MLE is unbiased.

36. Consider the density function

$$f(x) = e^{-(x-\alpha)}, \quad -\infty < \alpha \leq x.$$

(a) Find the maximum likelihood and method of moments estimators of α .

(b) Are both estimators found in (a) asymptotically unbiased?

Solution:

(a) To find the maximum likelihood estimator of α , find where the likelihood function is maximized.

$$\begin{aligned} f(x) &= e^{-(x-\alpha)} \quad -\infty < \alpha \leq x \\ L(\alpha|\mathbf{x}) &= \prod_{i=1}^n e^{-(x_i-\alpha)} \\ L(\alpha|\mathbf{x}) &= e^{-\sum_{i=1}^n (x_i-\alpha)} \end{aligned}$$

Thus, the largest value of α which yields non-zero answers is $\alpha = \min\{x_1, x_2, \dots, x_n\}$, so $\tilde{\alpha}(\mathbf{X}) = \min\{X_1, X_2, \dots, X_n\}$. To solve for the MOM estimator, $\alpha_1 = E[X] \stackrel{\text{set}}{=} \bar{X} = m_1$. Calculating $E[X]$:

$$\begin{aligned} E[X] &= \int_{\alpha}^{\infty} te^{-(t-\alpha)} dt \\ &= e^{\alpha} \int_{\alpha}^{\infty} te^{-t} dt \end{aligned}$$

Integrate by parts where $u = t$, $du = dt$, $dv = e^{-t}dt$, and $v = -e^{-t}$

$$\begin{aligned} &= e^{\alpha} \left[-te^{-t} \Big|_{\alpha}^{\infty} + \int_{\alpha}^{\infty} e^{-t} dt \right] \\ &= e^{\alpha} \left[\alpha e^{-\alpha} - e^{-t} \Big|_{\alpha}^{\infty} \right] \\ &= e^{\alpha} \left[\alpha e^{-\alpha} + e^{-\alpha} \right] \\ &= \alpha + 1 \end{aligned}$$

Thus, $\alpha + 1 \stackrel{\text{set}}{=} \bar{X}$ and the MOM of α is $\tilde{\alpha} = \bar{X} - 1$.

(b) The MLE must be asymptotically unbiased because ALL MLEs are asymptotically unbiased. The MOM is unbiased because $E[\bar{X} - 1] = E[X] - 1 = \alpha + 1 - 1 = \alpha$.